

Predictive Parsing (1)

Sam Ogden

DRAFT

This slide deck is still under active development. Expect changes before it is finalized.

Learning outcomes

- **After this lecture, you should be able to:**
- **write a “predictive parser” from a BNF grammar**
- **BNF**
- **define syntax**
- **write parser**
- **Parser**
- **code**

Warmup: deriving strings from BNF

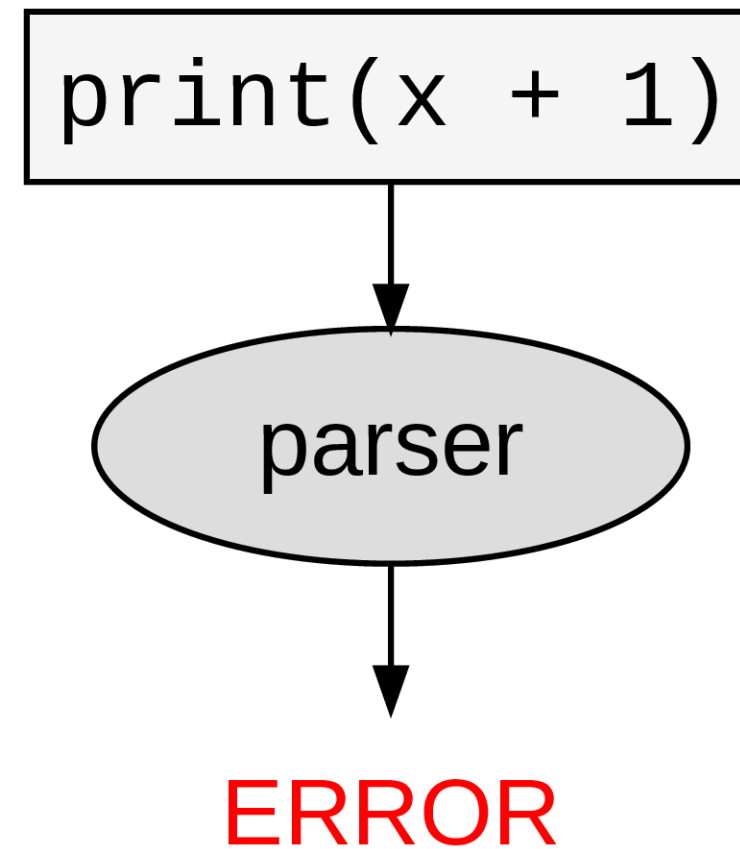
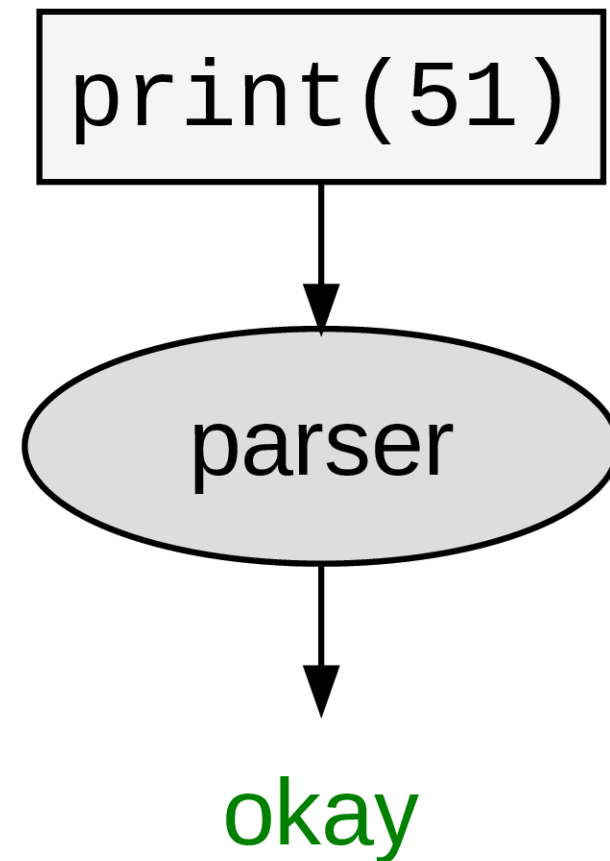
```
stmt ::= print(expr) | id=expr  
expr ::= id | num  
num ::= [0-9]+  
id ::= [a-zA-Z]+
```

Are these part of our language?

- `print(5)`
- `x = y`
- `x = print(5)`
- `print(x + 1)`

Recap: a parser checks syntax

```
stmt ::= print(expr) | id=expr  
expr ::= id | num  
num ::= [0-9]+  
id ::= [a-zA-Z]+
```



How to write parser for this grammar?

```
stmt ::= print(expr) | id=expr  
expr ::= id | num  
num ::= [0-9]+  
id ::= [a-zA-Z]+
```

Core Idea

- `stmt()` will parse statements
- `expr()` will parse expressions
- each function will decide which production to use by looking at the first symbol of each production
- once a production is picked the function will mimic the right-hand side of the production

Predictive parsing

- **Recursive-descent parsing: “top-down method of syntax analysis in which we execute a set of recursive procedures to process the input” (Aho et al, Dragon book)**
- **Predictive parsing: “a form of recursive-descent parsing in which the lookahead symbol determines the procedure selected for each non-terminal” (Aho et al, Dragon book)**

Main program

- initialize tokenizer
- call function associated with start symbol in grammar
- match on token **DONE** (end of input)

```
1 void parser() {  
2     Tokenizer t = init__tokenizer(str_to_parse);  
3     Stmt* e = parse_stmt(&t);  
4     if (is_done(t)) { printf("Success!"); }  
5 }
```


Example: function for 'stmt'

```
stmt ::= print(expr) | id=expr
expr ::= id | num
num ::= [0-9]+
id ::= [a-zA-Z]+
```

```
1 void stmt() {
2     switch (token) {
3         case PRINT:
4             match(PRINT); match("("); expr(); match(")");
5             break;
6         case ID:
7             match(ID); match("="); expr();
8             break;
9         default:
10            error("syntax error");
11    }
12 }
```

Challenge 1: Left recursion

- Any problems with a predictive parser here? $R1 ::= R1 + \text{term} \mid \text{term}$

Challenge 1: Left recursion

- Any problems with a predictive parser here? $R1 ::= R1 + \text{term} \mid \text{term}$

The BNF can be transformed into this:

```
R1 ::= termR2
R2 ::= +termR2 | ""
```

A general rule:

```
A ::= Aα | β
```

Derives the same strings as:

```
A ::= βR
R ::= αR | ""
```

α, β are sequences of **terminals** and **non-terminals** that don't start with **A**

Exercise

Rewrite to eliminate left recursion

```
R1 ::= R1+R1 | var
```

Exercise

Rewrite to eliminate left recursion

```
R1 ::= R1+R1 | var
```

Solution:

```
R1 ::= varR2  
R2 ::= +R1R2 | ""
```

Another solution:

```
R1 ::= varR2  
R2 ::= +R1R2 | ""
```

Challenge 2: Empty productions

```
R1 ::= var R2  
R2 ::= + R1 R2 | ""
```

How do we handle "" (i.e. the empty string, ϵ)?

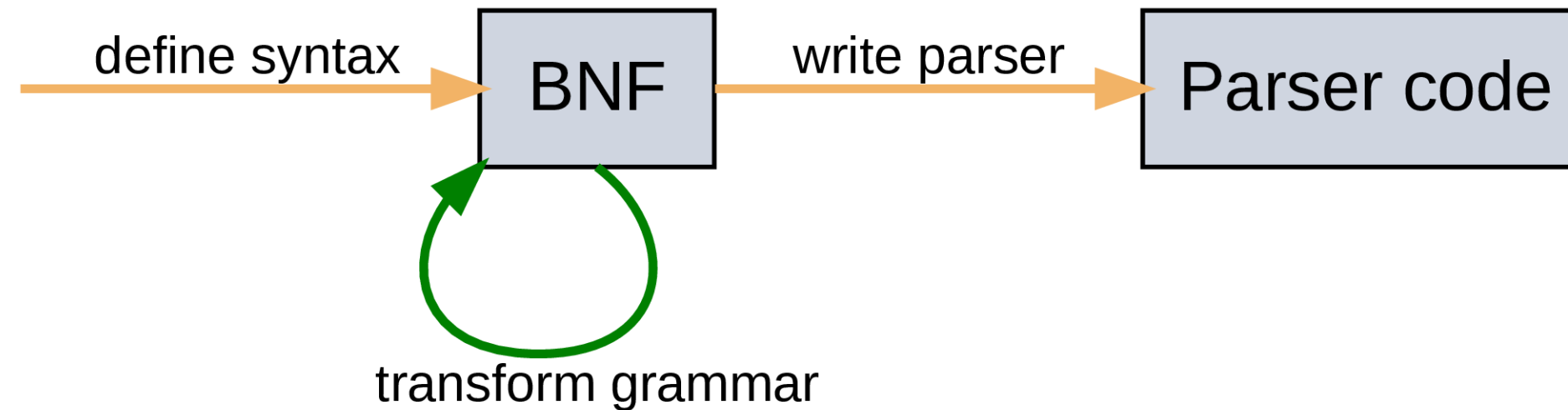
Challenge 2: Empty productions

```
R1 ::= var R2  
R2 ::= + R1 R2 | ""
```

How do we handle "" (i.e. the empty string, ϵ)?

```
1 void parse_R1() {  
2     switch (token) {  
3         case '+':  
4             match('+'); parse_R1(); parse_R2(); break;  
5         default:  
6             ; // empty production  
7     }  
8 }
```

Transforming a grammar



We transform the grammar in a way such that language of the grammar doesn't change.

Summary

- **predictive parsing is a simple kind of parser that is based directly on a BNF grammar**
- **but... you may need to modify a BNF grammar to allow predictive parsing to be used**